

Title

Comparable depression symptom improvement with donor versus autologous microbiota in bipolar patients

Abstract

Background: Treatment-resistant depression in bipolar disorder remains challenging. We evaluated microbiota manipulation as an adjunct to pharmacotherapy. **Methods:** 35 adults with bipolar disorder in depressive episodes were randomized to donor microbiota (n=17) or autologous microbiota (n=18) transplantation. All participants continued their established psychiatric medications. **Results:** Both treatment arms showed significant depression rating score improvements from baseline to week 24 (donor mean 16.7-point improvement vs autologous 15.4-point improvement). Statistical comparison revealed no significant difference between groups ($t=-0.47$, $p=0.64$, 95% CI -7.3 to 4.6). Microbial sequencing showed donor recipients achieved bacterial profiles resembling the healthy donor source with increased microbial diversity by 6 months, whereas autologous recipients showed minimal compositional change. No serious adverse events occurred in either group. **Conclusions:** Both FMT approaches yielded clinically meaningful symptom reduction, though microbial colonization differed. The absence of efficacy distinction between allogenic and autologous approaches suggests need for inert control in future studies.